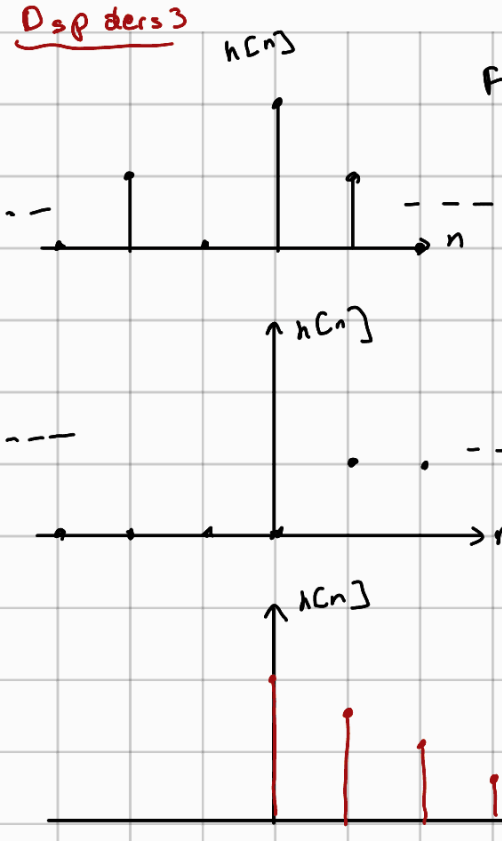
Dsp ders 3



FIR : sonlu sayıda 0'dan farklı değer alır

$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$

$x[n] \longrightarrow [\,h[n]\,] \longrightarrow y[n]$

$\delta[n] \rightarrow h[n]$ ve $\delta[n-k] \rightarrow h[n-k]$

$$y[n] = x[n] * h[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

$y[n]$ bunse — bu dense

$$= \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

olsa bir yerden sonra toplamı kesebilirdik

Filtre IIR se konvülasyon sadece kağıt üzerinde gerçekleşebilir yani toplam formülü kullanılamaz.

$$y[n] + \sum_{k=1}^{N} a_k\,y[n-k] = \sum_{k=0}^{M} b_k\,x[n-k]$$

$$y[n] = \sum_{k=0}^{M} b_k\,x[n-k] - \sum_{k=1}^{N} a_k\,y[n-k]$$

Constant coefficient Difference Equation

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] + \cdots + b_mx[n-m] - a_1y[n-1] - a_2y[n-2]$$

Sabit katsayılı fark denklemi

Bu nereden geliyor??

Birsten düşünelim

$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2] \longrightarrow h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

$$y[n-1] = \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2] + \frac{1}{3}x[n-3]$$

$$y[n] - y[n-1] = \frac{1}{3}x[n] - \frac{1}{3}x[n-3]$$

$$y[n] = \frac{1}{3}x[n] - \frac{1}{3}x[n-3] + y[n-1]$$

$b_0 = \frac{1}{3}$ $\quad a_1 = 1$

$b_1 = 0$ $\quad a_2 = 0$

$b_2 = 0$ $\quad a_3 = 0$

$b_3 = -\frac{1}{3}$

$b_4 = 0$

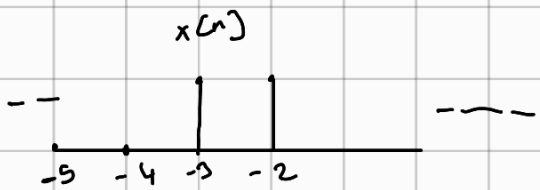
Örnek

$$y[n] = \frac{1}{100}x[n] + \frac{1}{100}x[n-1] + \cdots + \frac{1}{100}x[n-99]$$

$$y[n] - y[n-1] = \frac{1}{100}x[n] - \frac{1}{100}x[n-100]$$

$$y[n] = \frac{1}{100}x[n] - \frac{1}{100}x[n-100] + y[n-1]$$

$y[n] = 5x[n] + 3$ α lineer değil
LTI değil



Başlangıç koşulları

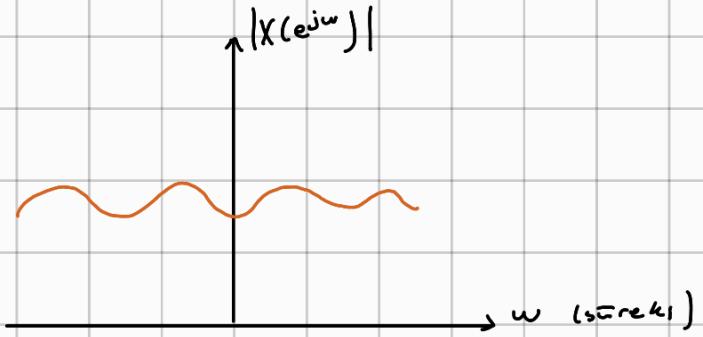
$y[-4] = 0$ $y[-6] = 0$

$y[-5] = 0$ $y[-1] = y[0] = 0$

DTFT → sonlu olmalı

$$X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n]e^{-jwn}$$

w'nin fonksiyonu sürekli



$\cos(-\Omega t)\big|_{t=nT}$
$t = \frac{n}{F_s}$

$\cos\left(\frac{\Omega}{F_s}n\right)$

$w = \frac{\Omega}{F_s}$

$\cos(wn)$

$-\infty < \Omega < \infty$

$-\infty < w < \infty$

$X(e^{jw})$ w'nin 2π sinde periyodik

$0 \le w < 2\pi$

$-\pi \le w < \pi$

(Ses işareti 3,2 | 4 kHz'e sınırlanıyor
8 kHz örnekleme frekansı)

$$X(e^{jw}) = X_R(e^{jw}) + jX_I(e^{jw})$$
$$= |X(e^{jw})|\, e^{j\angle X(e^{jw})}$$

↗ Fourier için yeter koşullar

$$|X(e^{jw})| = \left|\sum_{k=-\infty}^{\infty} x[k]e^{-jwk}\right| \le \sum_{k=-\infty}^{\infty} |x[k]e^{-jwk}| \le \sum_{n=-\infty}^{\infty} |x[k]| < \infty$$

↳ bu şartı sağlamayan

FIR sistemlerin her zaman fourieri

IIR sistemlerde sadece kararlıysa vardır

Properties of DTFT

$X(e^{jw}) = X(e^{jw} e^{j2\pi}) = Y(e^{j(w+2\pi)})$ Periyodik 2π ile

$x_1[n] \leftrightarrow X_1(e^{jw})$

$x_2[n] \leftrightarrow X_2(e^{jw})$

$a x_1[n] + b x_2[n] \leftrightarrow a X_1(e^{jw}) + b X_2(e^{jw})$

$\cos(wn) = \frac{1}{2} e^{jwn} + \frac{1}{2} e^{-jwn}$

$x[n] \xleftrightarrow{F} X(e^{jw})$

$x[n-n_0] \xleftrightarrow{F} e^{-jwn_0} X(e^{jw})$

$e^{jw_0 n} x(n) \xleftrightarrow{F} X(e^{j(w-w_0)})$

input ←

$Y(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$

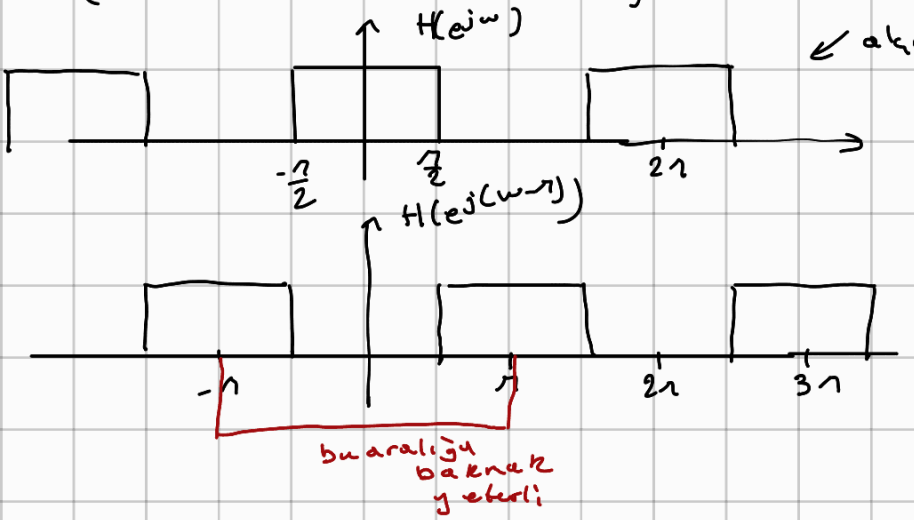
$X_1(e^{jw}) = \sum_{n=-\infty}^{\infty} e^{jw_0 n} x[n] e^{-jwn}$

$= \sum_{n=-\infty}^{\infty} x[n] e^{j(w-w_0)n}$

$w_0 = \pi \quad e^{j\pi n} = (-1)^n$

$h[n] \leftrightarrow H(e^{jw})$

$(-1)^n h[n] \leftrightarrow H(e^{j(w-\pi)})$



← alçak geçiren

bu aralığı bakmak yeterli

yüksek geçiren çünkü en yüksek frekans π !!

$x[n] \leftrightarrow X(e^{jw})$

$x^*[n] \leftrightarrow X^*(e^{-jw})$ ★ konjüge simetrik

$x[n]$ reel $\Rightarrow x[n] = x^*[n]$

$Y(e^{jw}) = X^*(e^{-jw}) \Rightarrow |X(e^{jw})| = |X^*(e^{-jw})|$

Fazlar orjine göre simetriktir

★ Reel bir işaretin fourier dönüşümü w'ya simetriktir

$x[n] \longleftrightarrow X(e^{j\omega})$

$x[-n] \longleftrightarrow X(e^{-j\omega})$

ispat ←

$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$X_1(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[-n] e^{-j\omega n}$

$X_1(e^{j\omega}) = \sum_{n=-\infty} x[k] e^{j\omega k}$

$\downarrow$

$e^{-j(-\omega)k}$

$x[n] - x[n-1] \xleftrightarrow{F} X(e^{j\omega}) - e^{-j\omega} X(e^{j\omega})$

türevlenebilir

$(1 - e^{-j\omega}) X(e^{j\omega})$



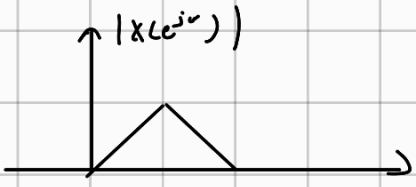
Yüksek frekanslar güçlenmiştir

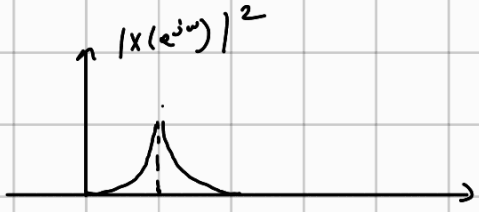
$n x[n] \hookrightarrow j \frac{d X(e^{j\omega})}{d\omega}$

$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$

$x[n]$ işaretinin enerjisi

Enerji yoğunluğu



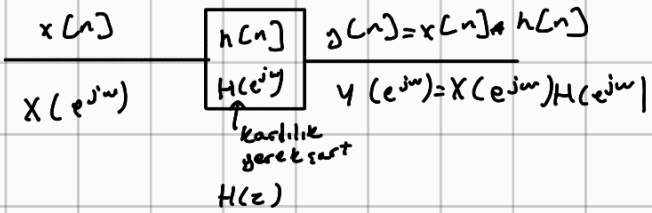


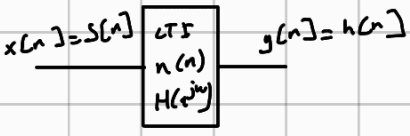
The convolution Theorem

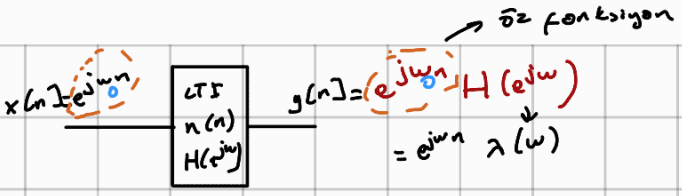
① $x_1[n] * x_2[n] \xleftrightarrow{F} X_1(e^{j\omega}) X_2(e^{j\omega})$

② $x_1[n] x_2[n] \xleftrightarrow{F} \frac{1}{2\pi} X_1(e^{j\omega}) X_2(e^{j\omega})$

$y[n] = x[n] * h[n] \xleftrightarrow{F} Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$







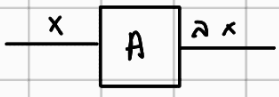
$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, e^{j\omega(n-k)}$$

$$= \left[\sum_{k=-\infty}^{\infty} h[k]\, e^{-j\omega k}\right] e^{j\omega n}$$

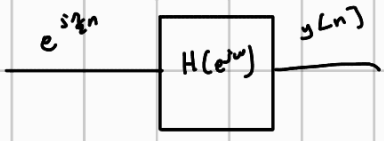
$$= e^{j\omega n}\, H(e^{j\omega})$$

öz vektör



$A\vec{x} = \lambda\vec{x}$

eigen value (öz değer) — eigen vector (öz vektör)



$y[n] = e^{j\frac{\pi}{2}n}\, H(e^{j\frac{\pi}{2}})$

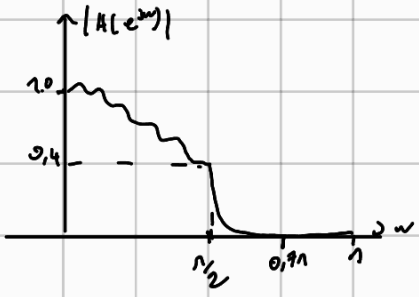
dizi — katsayı (örnek $0{,}4 \cdot e^{j\pi/4}$)

$\omega = \frac{\pi}{2}$ giriş için

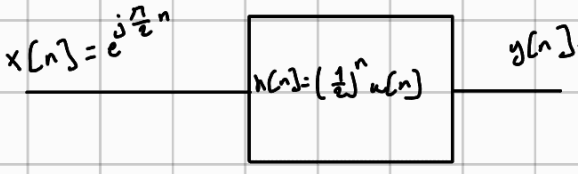
frekans değişmiyor

Lineer bir sistem giriş işaretin frekansı korunur.

fakat O'ra çevirebilir.



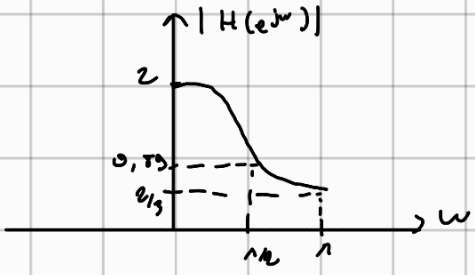
Girişin frekansına göre sistem farklı davranır bu yüzden $H(e^{j\omega})$'ya frekans cevabı denir.



$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n]\, e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \underbrace{\left(\frac{1}{2} e^{-j\omega}\right)^n}_{a} = \frac{1}{1 - \frac{1}{2} e^{-j\omega}}$$



$$y[n] = e^{j\pi/2 n}\, \underbrace{H(e^{j\pi/2})}_{0,89 e^{-j\,0,463}}$$

$$= 0,89\, e^{j(\frac{\pi}{2} n - 0,463)}$$

